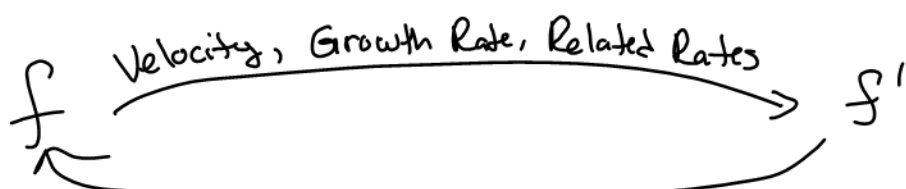


Lecture 16: Relative Extrema and Critical Numbers

Goal: Discuss when a function's behavior changes.



Extrema, Intervals of Inc./Dec., Concavity, Inflection Points

Consider the following graph:

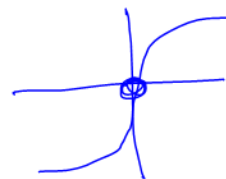
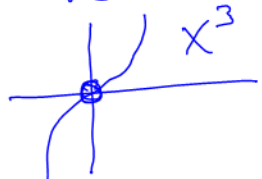


Def A fun f has a relative maximum (or local max) at $x=c$ if there is an interval I containing c such that $f(x) \leq f(c)$ for x in I .

Def A relative minimum (or local min) is obtained by replacing " $f(x) \leq f(c)$ " with " $f(c) \leq f(x)$ " is the prior def.

Theorem (Fermat, ~1636) If f has a rel. max or min at a c , then $f'(c)$ is either 0 or undefined.

NOTE: The reverse is not true $x^{\frac{1}{3}}$



Def Let c be in the domain of f . If $f'(c)$ is 0 or undefined, we say c is a Critical number (or critical point or critical value) of f .

Example 1 Let $f(x) = 2\cos(5x) + 5x$
@ Is $x = \frac{\pi}{30}$ a critical number for f ?

$$f'(x) = -10\sin(5x) + 5$$

$$f'\left(\frac{\pi}{30}\right) = -10\sin\left(5 \cdot \frac{\pi}{30}\right) + 5 = -10\sin\left(\frac{\pi}{6}\right) + 5$$
$$= -10\left(\frac{1}{2}\right) + 5 = 0 \quad \checkmark \text{ Yes}$$



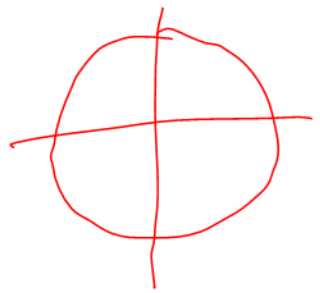
(b) Is $x = \pi$ a crit. num. for f ?

$$f'(x) = -10\sin(5x) + 5$$

$$f'(\pi) = -10\sin(5\pi) + 5$$

$$= 5 \neq 0 \text{ or undefined}$$

X; it is not a crit. num.



Example 2 Find all critical numbers for

$$f(x) = x^3 - 75x$$

$$f'(x) = 3x^2 - 75$$

f' is never undefined. So, we need to solve

$$3x^2 - 75 \stackrel{\text{set}}{=} 0$$

$$3(x^2 - 25) = 0$$

$$3(x-5)(x+5) = 0$$

$$\Rightarrow x = -5, 5$$

Example 3 Find all critical numbers for

$$f(x) = x^3 + 5x^2 + 7x$$

$$f'(x) = 3x^2 + 10x + 7$$

f' is never undefined. So, we do the following

$$3x^2 + 10x + 7 \stackrel{\text{set}}{=} 0$$

$$ax^2 + bx + c = 0 \Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-10 \pm \sqrt{(10)^2 - 4(3)(7)}}{2(3)} = \frac{-10 \pm \sqrt{100 - 84}}{6}$$

$$= \frac{-10 \pm \sqrt{16}}{6} = \frac{-10 \pm 4}{6} \Rightarrow x = \frac{-10+4}{6}, x = \frac{-10-4}{6}$$

$$\Rightarrow x = -1, -\frac{7}{3}$$

Example 4 Find all critical numbers of

$$f(x) = 4x^3 - \frac{4}{x^3}$$

$$\begin{aligned} f'(x) &= 12x^2 - [4x^{-3}]' = 12x^2 + 12x^{-4} \\ &= 12x^2 + \frac{12}{x^4} \end{aligned}$$

NOTE: It is tempting to say 0 is a critical number here since $f'(0)$ is undefined. But remember, critical numbers are in the domain of the function, which 0 is not.

f' is never undefined (in the domain of f).

So, we solve $f'(x) = 0$

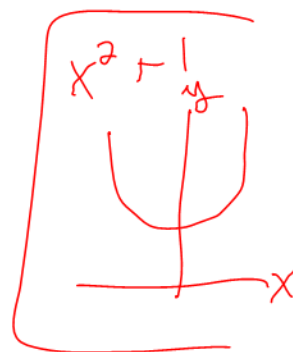
$$12x^2 + \frac{12}{x^4} \stackrel{\text{set}}{=} 0$$

$$x^6 \cdot x^4 \left(12x^2 + \frac{12}{x^4} \right) = 0 \cdot x^4$$

$$12x^6 + 12 = 0$$

$$x^6 + 1 = 0$$

NO Real Solution



$\Rightarrow f$ has no critical numbers.

Example 5 Repeat for

$$f(x) = xe^x$$

$$f'(x) = (1) e^x + x e^x$$

f' is never undefined. So, we solve

$$1 \cdot e^x + x e^x \stackrel{\text{set}}{=} 0$$

$$e^x (1+x) = 0$$

$$e^x = 0$$

Never

OR

$$1+x=0$$

$$\boxed{x = -1}$$

Example 6 Repeat for

$$f(x) = 6x^4 e^{5x+4}$$

$$f'(x) = 24x^3 e^{5x+4} + 6x^4 [5 \cdot e^{5x+4}]$$

$$= 24x^3 e^{5x+4} + 30x^4 e^{5x+4} \stackrel{\text{set}}{=} 0$$

$$6x^3 e^{5x+4} [4 + 5x] = 0$$

$$6x^3 = 0 \quad \text{OR} \quad e^{5x+4} = 0 \quad \text{OR} \quad 4+5x=0$$

$$\boxed{x=0} \quad \text{Never} \quad \boxed{x = -\frac{4}{5}}$$

Ex 7 Repeat for $f(x) = \frac{\ln x}{x}$

$$f'(x) = \frac{\frac{1}{x} \cdot x - (1) \ln x}{x^2} = \frac{1 - \ln x}{x^2}$$

$f'(x)$ is never undefined (in the domain of f)

$$\frac{1 - \ln x}{x^2} = 0$$

$$1 - \ln x = 0$$

$$\ln x = 1$$

$$\boxed{x = e}$$

